Constellation

10 CFR 50.73

CCN: 25-86

October 8, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Units 2 and 3
Subsequent Renewed Facility Operating License No. DPR-44, DPR-56
NRC Docket No. 50-277, 50-278

Subject: Licensee Event Report (LER) 2025-002-00 Uncoordinated DC Circuits Cause Unanalyzed Condition

Reference: ENS 57899

The subject report is being submitted in accordance with 50.73(a)(2)(ii)(B) for being in an unanalyzed condition that significantly degraded plant safety, due to uncoordinated DC circuits that result in being outside the bounds of the 10CFR50 Appendix R Fire Safe Shutdown Analysis.

There are no commitments contained in this letter. If you have any questions, please contact the Peach Bottom Regulatory Assurance Manager, Mr. Brian Woodard at (257) 533-7430.

Respectfully,

Stiltner, Ryan C

Digitally signed by Stiltner, Ryan C
Date: 2025.10.08 11:54:47 -04'00'

Ryan C. Stiltner
Plant Manager
Peach Bottom Atomic Power Station

Enclosure

cc: USNRC, Administrator, Region I
USNRC, Senior Resident Inspector
W. DeHaas, Commonwealth of Pennsylvania
S. Seaman, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name: Peach Bottom Atomic Power Station (PBAPS), Unit 2

☒ 050 ☐ 052 **2. Docket Number:** 00277

4. Title: Uncoordinated DC Circuits Cause Unanalyzed Condition

5. Event Date			6. LER Number			7. Report Date		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year
09	04	2025	2025	- 002 -	00	10	08	2025

8. Other Facilities Involved

Facility Name		Docket Number
PBAPS Unit 3	☒ 050	00278
	☐ 052	

9. Operating Mode: 1

10. Power Level: 100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Brian Woodard, Regulatory Assurance Manager	(267) 533-7430

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	EI	FU	R352	Y					

14. Supplemental Report Expected: ☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date: Month | Day | Year (blank)

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 09/04/2025, an analysis was completed which confirmed that six cables associated with direct current circuits (EIIS:EI) at PBAPS Units 2 and 3 were non-compliant with 10CFR50 Appendix R, Section III.G.1, Fire protection of safe shutdown capability. It was determined that the circuits were uncoordinated due to inadequate fuse sizing, meaning that during a postulated fire, the circuit's fuse may not act to protect the circuit from a fire-induced hot short. Without adequate overcurrent protection for these circuits, a fire-induced short could lead to excessive current through the circuits. This could lead to secondary damage in another fire area where the circuits are routed, challenging the ability to achieve and maintain safe shutdown. Compensatory measures have been implemented and will remain until the identified non-compliant conditions are resolved.

This event is reported under 10CFR50.73(a)(2)(ii)(B) as an unanalyzed condition because a fire that affects multiple fire areas is outside the bounds of the 10CFR50 Appendix R Fire Safe Shutdown Analysis.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Peach Bottom Atomic Power Station (PBAPS), Unit 2	■ 050 / ☐ 052	00277	2025	- 002	- 00

NARRATIVE

Plant Operating Conditions Before the Event
Prior to the event, Peach Bottom Atomic Power Station (PBAPS), Unit 2 was operating at approximately 100% power in MODE 1. PBAPS, Unit 3 was operating at approximately 96% power in MODE 1. There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

Event Description
On 09/04/2025, an analysis was completed which confirmed that six cables associated with direct current circuits (EIIS:EI) at PBAPS, Units 2 and 3 were non-compliant with 10 CFR 50 Appendix R, Section III.G.2,Fire protection of safe shutdown capability. It was determined that the circuits were uncoordinated due to inadequate fuse sizing, meaning that during a postulated fire, the circuit's fuse may not act to protect the circuit from a fire-induced hot short. Without adequate overcurrent protection for these circuits, a fire-induced short could lead to excessive current through the circuits. This could lead to secondary damage in another fire area where the circuits are routed, challenging the ability to achieve and maintain safe shutdown.

Compensatory actions were implemented immediately for the affected fire areas in accordance with procedural requirements. Compensatory actions will remain in place until the identified non-compliant conditions are resolved.

The circuits for PBAPS,Units 2 and 3 noted above were reported under ENS 57899 per 10CFR50.72(b)(3)(ii)(B) as an unanalyzed condition because a fire that affects multiple fire areas is outside the bounds of the 10CFR50 Appendix R Fire Safe Shutdown Analysis.

Safety Consequences
There were no actual consequences caused by this condition. The potential consequence of a hot short condition is for fire damage in one fire area to occur which subsequently damages cables in a second fire area. This chain of events could cause a loss of safe shutdown capability outside the bounds of the Fire Safe Shutdown Analysis. However, the risk of fire damage is limited by the use of fire retardant cabling, divisional separation, and fire protection features such as automatic suppression and detection, where required. Additionally, the station Fire Brigade is trained and will readily respond to all fire conditions.

Cause and Corrective Actions
The cause is an inadequate design dating back to original construction. Fire Protection Program compensatory measures were implemented as immediate actions to mitigate the risk of the affected circuits. Corrective Actions will include modification of the affected circuits to ensure they are properly coordinated.

Previous Similar Events
Peach Bottom LER 2-18-001, dated 05/18/2018, reported an unanalyzed condition due to valve cables not being protected from spurious operation during a postulated fire.

Peach Bottom LER 2-14-001, dated 07/17/2014, reported an unanalyzed condition due to broken wires in breakers credited in the Appendix R Post-Fire Safe Shutdown Analysis.

Peach Bottom LER 3-11-04, dated 01/13/2012, reported an unanalyzed condition due to an error in cable routing which affected the Post-Fire Safe Shutdown Analysis.